

DS 4420 PROJECT — Game Recommendation System

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PURPOSE

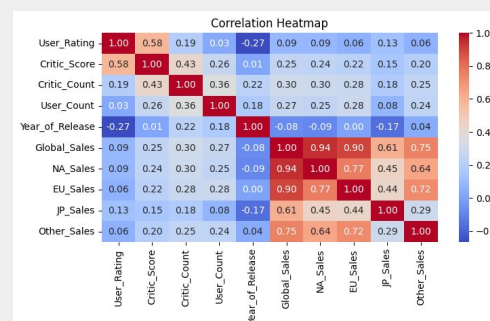
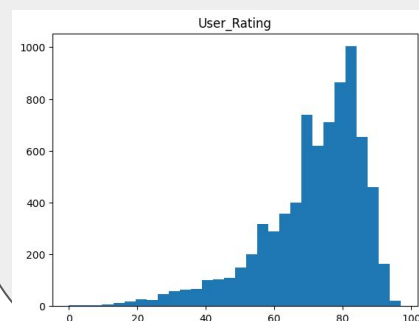
- Video game user ratings play a key role in what games players buy and what studios choose to fund
- To predict user rating(0-100) from available information such as platform, genre, ESRB rating, release year, critic score, critic count and sales
- To identify which game characteristics are most strongly associated with highly rated game

Background & Data

Video game review strongly influence what player will buy and which project should the studio fund. User rating is one of the most significant attribute and summarize how satisfied players are after buying the game. However, using rating are not published until the game releases so have to find a way to predict them from pre-release information, which is useful for both publishers and recommendation systems.

To study this problem, we used a curated version of Video Game Sales with Ratings dataset. The dataset contains commercial games released on major platforms prior to 2016 and combines player ratings, critic reviews, and global sales information.

Each game has a user rating on a 0–10 scale that we rescale to 0–100 to match critic scores. For predictors we include platform (e.g., PlayStation, Xbox, PC, Nintendo), genre, ESRB rating, and year of release, along with critic score, critic review count, user review count, and both regional and global sales. We restrict our analysis to games with non-missing user ratings and complete information on these key variables, and we standardize continuous features (scores, counts, sales, year) so that they are on comparable scales for both the wide-and-deep MLP and the Bayesian regression model.



METHODS

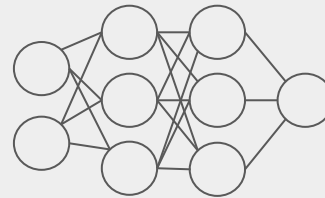
MLP - Deep & Wide

Architecture:

- Wide: Linear layer on Platform × Genre cross-features (memorization)
- Deep: 2-layer neural network on standardized features (generalization)
- Final pred: sum of wide + deep outputs

Implementation:

- Manual backpropagation (NumPy only)
- ReLU activations
- Mini-batch gradient descent
- L2 regularization



Bayesian Regression

Formulation:

- Gaussian likelihood on User_Rating
- Normal priors on coefficients
- Student-t prior on residual σ
- MCMC inference (using brms)

Key Features:

- Interpretable coefficients
- 95% credible intervals
- Uncertainty quantification
- Grouped rare categories (<100 obs)

Likelihood:

$$y_i | \beta, \sigma^2 \sim \mathcal{N}(\mathbf{x}_i^T \beta, \sigma^2)$$

Priors:

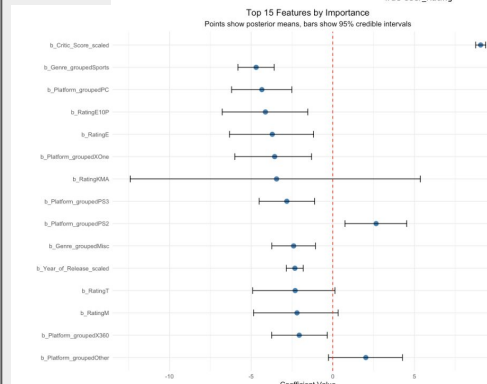
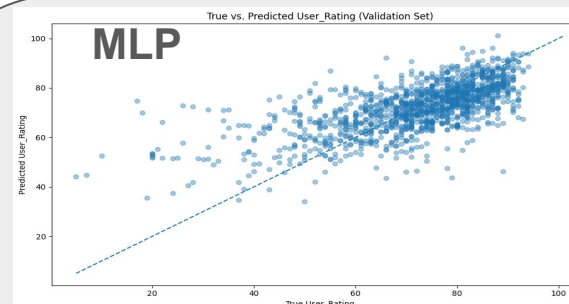
$$\beta \sim \mathcal{N}(0, 5)$$

$$\sigma \sim \text{Student-}t(3, 0, 10)$$

CONCLUSION

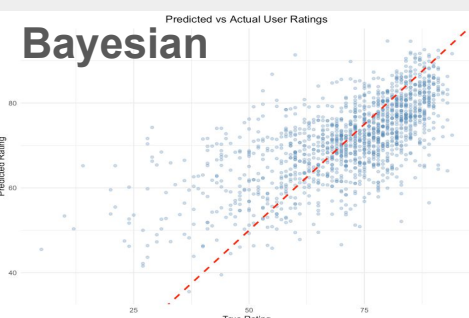
- Using game metadata, critic reviews, and sales, both models substantially outperform a naive baseline that always predicts the mean user rating (baseline RMSE $\approx 14-15$; MLP ≈ 10.6 ; Bayesian ≈ 10.9 , MAE ≈ 8.1)
- A relatively small, interpretable set of features - platform, genre, ESRB rating, critic score, user count, release year, and sales - capture most of the variation in user ratings across games.
- The wide and deep MLP provides slightly better predictive accuracy, and Bayesian regression offer clear interpretation and uncertain intervals for each prediction

RESULTS



Model	RMSE	Improvement
Baseline (mean)	14.7	—
Wide & Deep MLP	10.6	28%
Bayesian Regression	10.9	26%

Both models significantly outperform baseline. MLP achieved better accuracy, but Bayesian provides interpretable coefficients and uncertainty estimates.



Critic Score is the strongest predictor (+9.05 coefficient)

Sports games rated 4.7 points lower on average

Older games rated higher (survivorship bias)

Platform matters: PS2 (+2.7), PC (-4.3), Xbox One (-3.6)

95% credible intervals capture most true ratings

Future Work

- Add interaction terms (Platform X Genre effects)
- Compare to collaborative filtering approaches
- Incorporate review text using NLP (natural language processing)
- Hierarchical Bayesian model with varying effects
- Test further on post 2016 games to validate generalization
- Build user-level recommendation system

REFERENCE

Cheng, H. T., et al. (2016). Wide & Deep Learning for Recommender Systems. ArXiv.

Lim, Y. J., & Teh, Y. W. (2007). Variational Bayesian Approach to Movie Rating Prediction.

Sifa, R., Drachen, A., & Bauckhage, C. (2021). Large-Scale Cross-Game Player Behavior Analysis on Steam. AIIDE.

Dataset: rush4ratio (2016). Video Game Sales with Ratings. Kaggle.